10(a)(i)	to increase the magnetic flux linkage (between the coils)	**B1**
10(a)(ii)	to reduce energy losses	**B1**
	by reducing induced currents	**B1**
10(b)(i)	maximum V_{OUT} = 12 000 × (625 / 25 000) = 300 V	**A1**
10(b)(ii)	r.m.s. current = 300 / (640 × √2) = 0.33 A	**A1**
10(b)(iii)	sketch: sinusoidal shape in positive half of the graph, sitting with ‘minima’ resting on the time-axis (at *P* = 0)	**B1**
	each ‘cycle’ shown repeating every 20 ms	**B1**
	maximum *P* shown as 140 W	**B1**
10(c)	power curve is symmetrical about the midpoint (on the power axis)	**B1**
	mean power is half the peak power	**B1**